

The $d = 2$ cutoff expansion with unequal starting exponents

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Abstract

Astra #8 rank 3 (abstract half), guided by Astra consult #9. The equal-exponent assembly used $(h_1 + 1)/k_1 = (h_2 + 1)/k_2$ only for exponent bookkeeping. With $\alpha_i = (h_1 + i + 1)/k_1$, $\delta_j = (h_2 + j + 1)/k_2$ and $D = \delta_j - \alpha_i$ the face-vs-corner cancellations are identities in D alone (`uface_coeff_eq_corner_fp'`), so the over-complete sum equals the merged sum for any corner order (`overcomplete_eq_merged`); the u -face remainder exponent is $p_2 + M_2/k_2$, the v -face one $p_1 + M_1/k_1$, the mixed one $E = \min$ of the two; the shifted compatibilities $p_1 + (M_1 - 1)/k_1 < p_2 + M_2/k_2$ (and symmetrically) give the face admissibilities and, as Astra #9 observed, also the corner admissibility $\gamma_{ij} < M_1$, so the old corner order $M_1 + k_1 M_2$ survives. Result: `twoD_cutoff_general_merged`, the cutoff expansion of the $d = 2$ block with independent starting exponents and remainder $O(N^{-E}(1 + \log N))$. Zero `sorry/axiom`.

1 The things formalised

```
theorem sub_gamma_u (h h k k i j : ) (hk : 0 < k) :
  (j : ) - ((k : ) * (((h + i : ) : ) + 1) / k) - h - 1)
  = (k : ) * (((h + j : ) : ) + 1) / k - ((h + i : ) : ) + 1) / k)
theorem uface_coeff_eq_corner_fp' (b : ) (h h k k i j Mc : ) (hb : 0 < b)
  (hk : 0 < k) (hk : 0 < k) (hMc : 0 < Mc) (c : → → → ) :
  faceCoeff ((k : ) * (((h + i : ) : ) + 1) / k) - h - 1) (b ^ k) k k
    (fun m => c i m) j
  = faceFPCoeff ((k : ) * (((h + j : ) : ) + 1) / k) - ((h + i : ) : ) - 1)
    b k (fun _ s => c i j s) (cornerData (c i j)) Mc
theorem overcomplete_eq_merged ( b : ) (h h k k M M Mc : ) (hb : 0 < b)
  (hk : 0 < k) (hk : 0 < k) (hMc : 0 < Mc) (a bj : → → → )
  (c : → → → ) (N : ) :
  (u-face sums, order M) + (v-face sums, order M) - (corner sums, order Mc)
    = mergedSum b h h k k M M a bj c N
theorem corner_gamma_lt_general (p p : ) (h h k k M M : ) (hk : 0 < k)
  (hk : 0 < k) (hp : ((h : ) + 1) / k = p) (hp : ((h : ) + 1) / k = p)
  (hM : p + ((M : ) - 1) / k < p + (M : ) / k) (i j : ) (hj : j < M) :
  (k : ) * (((h + j : ) : ) + 1) / k) - ((h + i : ) : ) - 1 < M
theorem twoD_cutoff_general_merged ( b p p : ) (h h k k M M : ) (h : 0 < )
  (hb : 0 < b) (hk : 0 < k) (hk : 0 < k) (hp : ((h : ) + 1) / k = p)
  (hp : ((h : ) + 1) / k = p)
  (hM : p + ((M : ) - 1) / k < p + (M : ) / k)
  (hM : p + ((M : ) - 1) / k < p + (M : ) / k)
  ... (same amplitude, face, remainder, envelope and moment hypotheses as before) :
  (fun N : => twoDAMP b N h h k k Φ - mergedSum b h h k k M M a bj c N)
    = 0[atTop] fun N : =>
      N ^ (-(min (p + (M : ) / k) (p + (M : ) / k))) * (1 + Real.log N)
```

2 Mathematics

The u -face i expansion has transverse parameter $\gamma_i = k_2\alpha_i - h_2 - 1 = k_2(\alpha_i - p_2)$; its monomial $m = j$ sits at $\alpha_i + (j - \gamma_i)/k_2 = \delta_j$ and its coefficient $-(k_1)^{-1}(b^{k_1})^{-(j-\gamma_i)/k_2} c_{ij}/(j - \gamma_i) = -b^{-k_1 D} c_{ij}/(k_1 k_2 D)$ equals the corner's finite-part coefficient $k_2^{-1} c_{ij} b^{-\gamma_c}/(-\gamma_c)$ with $\gamma_c = k_1(\delta_j - \alpha_i) = k_1 D$; at a collision both are $c_{ij} \log b/k_2$. Nothing here refers to $p_1 = p_2$. The remainder exponents: $\alpha_i + (M_2 - \gamma_i)/k_2 = p_2 + M_2/k_2$ and $\delta_j + (M_1 - \gamma'_j)/k_1 = p_1 + M_1/k_1$, both $\geq E$; the mixed remainder is exactly E . The admissibility $\gamma_i < M_2$ is $k_2(\alpha_i - p_2) < M_2$, i.e. $\alpha_i < p_2 + M_2/k_2$, which the first shifted compatibility gives for all $i < M_1$. For the corners, $\gamma_{ij} = k_1(\delta_j - p_1) - i$ and the second compatibility gives $\delta_j < p_1 + M_1/k_1$ for $j < M_2$, hence $\gamma_{ij} < M_1 \leq M_1 + k_1 M_2$, and the corner remainder exponent $\alpha_i + (M_1 + k_1 M_2)/k_1 \geq p_1 + M_1/k_1 \geq E$.

3 Paper-facing meaning

This removes the equal-exponent restriction from the $d = 2$ block: the cutoff expansion now covers every pair (h, k) , with the leading candidate exponent $\min(p_1, p_2)$ and a leading logarithm possible only when $p_1 = p_2$ (later logarithms still occur at collisions $\alpha_i = \delta_j$). Astra #9's answers on the rest of the programme: the leading coefficient for $p_1 < p_2$ is $B_{p_1} = k_1^{-1} \sum_j \frac{b^{j+\Delta}}{j+\Delta} M[p_1; 0; c_{0j}]$ with $\Delta = k_2(p_2 - p_1)$ (a genuine integral, no counterterms); next the analytic adapter and the general Taylor tree (cutoffs $q = \max(2T, p_1, p_2) + 1$, $M_\ell = \lceil k_\ell(q - p_\ell) \rceil$), then the equal theorem as a corollary; afterwards (R5) an explicit-constant cutoff bound and uniform families, (R6) coefficient maps as bounded linear maps in a weighted norm with $r < \rho$, (R7) Taylor-data identification by one-variable differentiation twice, (R9) expectation commuting with coefficient extraction under an integrable majorant, (R8) the general- d constant product density.

4 Lean notes

- `push_cast` zeta-unfolds `set` variables in the goal, so hypotheses stated with the abbreviation no longer match for `linarith`; prove the needed real identities as `haves` in terms of the abbreviations and avoid `push_cast` on goals mentioning them.
- Keep two forms of a cast identity: `(0 : ) - c` for `linarith` and `((0 : ) : ) - c` for rewriting the `axisPrim` branch (`Nat.cast_zero` bridges them).
- From $kD = 0$ with $k \neq 0$, `(mul_eq_zero.1 h).resolve_left` gives $D = 0$; then `rw [hD0, mul_zero]` at `hu` before `linarith` (it cannot multiply hypotheses).
- `field_simp` may close a goal completely; a trailing `ring` then fails with “no goals”.
- When a lemma's statement carries $\uparrow(h + j)$ and the goal has been `push_cast`, apply `push_cast at h` to both.
- The degenerate case $M_1 = 0 \vee M_2 = 0$ of the merged form needs a separate branch since the corner order must be positive in `overcomplete_eq_merged`; there all corner sums vanish.